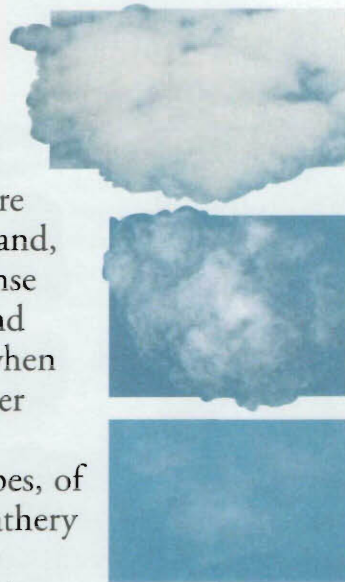
CLOUDS

WHEN YOU LOOK UP at the sky, you may see clouds. In temperate or mild climates, there are usually at least a few clouds and, sometimes, cloud cover is total. Clouds are dense masses of water drops or ice crystals so light and small that they float on the air. Clouds form when rising air cools to a point where it can no longer contain its water vapour, and so the vapour condenses. There are three basic forms, or shapes, of cloud – puffy cumulus, layered stratus, and feathery cirrus – but each form can vary to make many different cloud types. The type of cloud depends on how high the air rises, and its temperature.



Cloud formation

Clouds form by the condensation or freezing of water vapour. The way they form depends on their height and on the speed of upward air movement. When pockets of warm air rise rapidly, clouds form in heaped shapes (cumulus). When air rises slowly and evenly over a large area, clouds form in layers (stratus).

Making a cumulus
The sun-warmed ground creates thermals – rising currents of warm air. The air cools as it rises. Eventually, it becomes so cool that water droplets condense and a cloud forms. The cloud continues to build up as long as the thermal continues to supply water vapour.

Formation of a cumulus cloud in three stages

Cirrus clouds form at high altitude where air is cold and strong.

Cirrostratus is a high level veil of cirrus cloud.

Altostratus is a thin watery sheet of cloud.

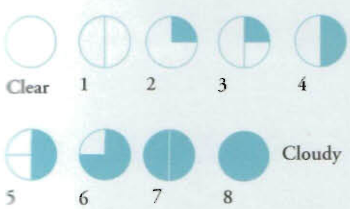
Cirrocumulus are clouds of ice crystals with a dapple appearance.

Luke Howard

A keen amateur meteorologist, but a pharmacist by profession, British-born Luke Howard (1772–1864) kept detailed weather diaries. These provided valuable meteorological data, before official records were kept. Howard used Latin names to identify each cloud by shape. His classification of clouds is still used today.

Cloud types

There are 10 distinct types of cloud. Cirrus, cirrostratus, and cirrocumulus clouds form 5–11 km (3–7 miles) above sea-level. Altocumulus, altostratus, and nimbostratus clouds form 2–7 km (1–4 miles) above sea-level. Stratocumulus and stratus form at 2 km (1 mile) or under above sea-level. Cumulus and cumulonimbus clouds form over a wide range of heights.



Cloud cover
The amount of sunlight reaching the ground depends on how much sky is covered by cloud. This is measured in "oktas" (eighths). One okta means one-eighth of the sky is covered in cloud; two oktas equals two-eighths of cloud cover in the sky, and so on.

Fog and mist

When water vapour in the air condenses near the ground it forms fog and mist. "Radiation" fog forms on cold, clear, calm nights, when the ground rapidly loses the heat it has absorbed during the day and cools the air above to its dew point. "Advection" fog forms when warm, moist air flows over a surface so cold that the water vapour in the air condenses.

Sea mist
When warm moist air flows over cold water, water vapour in the air may condense to form a kind of advection fog called a sea mist. These mists are most common on early summer mornings, when the air is calm.

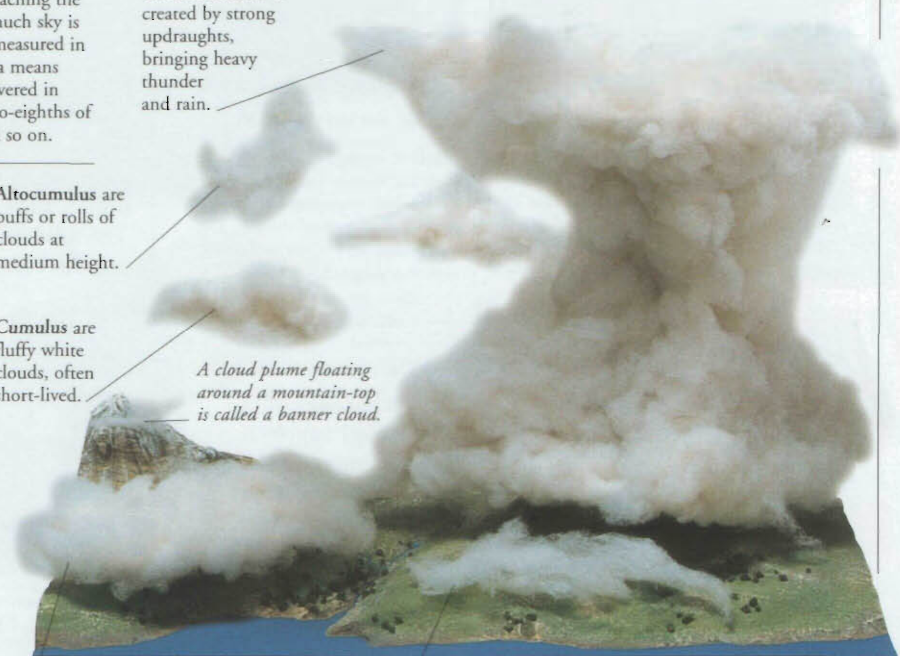
Beachy Head, Sussex, England

Cumulonimbus is created by strong updraughts, bringing heavy thunder and rain.

Altocumulus are puffs or rolls of clouds at medium height.

Cumulus are fluffy white clouds, often short-lived.

A cloud plume floating around a mountain-top is called a banner cloud.



Nimbostratus are layers of dark rain clouds.

Stratus are cloud layers, often giving long periods of rain.

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